

Evanns Morales-Cuadrado

U.S. Citizen

Robotics PhD Student Georgia Institute of Technology

CONTACT

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Github: <https://github.com/evannsmc>

Research Website: <https://www.evannsmc.com/>

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TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Matlab/Simulink
- **Frameworks & Libraries:** JAX, PyTorch, acados, CasADi, ROS 2, PX4, ArduPilot
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native)

EDUCATION

Georgia Institute of Technology | 2022-2027

Ph.D. in Robotics

Doctoral Minor in Optimization Methods

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington | 2018-2022

Honors Bachelor of Science in Electrical Engineering

Minor in Mathematics

Minor in Physics

Certificate in Unmanned Vehicle Systems

- GPA: 3.9/4.0 Summa Cum laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

WORK EXPERIENCE

Graduate Researcher | AI for Autonomy

May 2026 - Present

Sandia National Laboratories

Albuquerque, NM

- Conducted applied research integrating artificial intelligence and advanced control methods for autonomous systems
- Developed and evaluated algorithms for the perception, decision-making, and control of autonomous platforms
- Implemented and tested methods in simulation and on hardware within a multidisciplinary research team

Graduate Research Assistant | FACTS Lab

August 2022 - Present

Georgia Institute of Technology

Atlanta, GA

- Advised by Dr. Samuel Coogan
- Implemented novel aggressive tracking control methods on quadrotor hardware
- Developed novel nonlinear control methods for safe autonomy
- Published novel research at top conferences and journals in the fields of robotics and control theory

Research Assistant | Autonomous Systems Lab

November 2021 - May 2022

University of Texas at Arlington

Arlington, TX

- Advised by Dr. Frank Lewis
- Received personal mentorship from a leading researcher in the field of controls and reinforcement learning (ranked 12th in the USA and 23rd in the world during my time in his lab)
- Aided graduate students on reinforcement learning research applied to quadrotor control

Research Assistant | Dynamical Networks and Control Lab
University of Texas at Arlington

October 2019 - May 2022
Arlington, TX

- Advised by Dr. Yan Wan
- Gained experience with Robot Operating System (ROS), OpenCV, and path planning
- Original research in learning-based minimum time and energy path-planning for multi-vehicle systems

**SELECTED
PUBLICATIONS**

E. Morales-Cuadrado, L. Chung, S. Kousik and S. Coogan, “RTD-RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances.” Joint Submission to 2026 MECC-ALDSC. (*under revision*) [preprint available](#).

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, “Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic.” Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (*under revision*) [preprint available](#).

E. Morales-Cuadrado, C. Llanes, Y. Wardi and S. Coogan, “Newton-Raphson Flow for Aggressive Quadrotor Tracking Control.” [2024 American Control Conference \(ACC\)](#)

**RESEARCH
INTERESTS**

- Safe Autonomy
- Hardware Deployment
- Unmanned Ground and Aerial Vehicles
- Advanced Nonlinear Control
- Learning-Based Control
- Trajectory Generation and Planning

**OPEN-SOURCE
CONTRIBUTIONS**

- **PX4 Autopilot:** Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4:** Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech’s Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

**SERVICE &
LEADERSHIP**

Co-Founder and President of Puerto Rican Student Association
Georgia Institute of Technology

August 2024 - Present
Atlanta, GA

- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
 - Formed a three-member co-founder board and identified a faculty advisor
 - Raised funds for the organization and recruited members from campus
 - Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech
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HONORS AND AWARDS	Top-3 Finalist at a Deep Learning Research Symposium	November 2023
	Professor-sponsored award for novel research in deep learning and robotics	
	Goizueta Fellowship at Georgia Tech	August 2022
	Highly selective fellowship for graduate students of Latin-American descent	
	Summa Cum Laude Honors at University of Texas at Arlington	May 2022
	Exceptional academic distinction in top GPA tier for graduation honors	
	Electrical Engineering Honors Scholar at University of Texas at Arlington	December 2021
	Inaugural member of the electrical engineering honors cohort	
	IEEE-HKN International Electrical Engineering Honor Society Membership	August 2021
	Selective undergraduate society recognizing outstanding academic achievement and community service	
	Chance Vought Engineering and Science Endowment Scholarship	August 2020
	Highly selective award given annually to three recipients across the entire College of Engineering	
	National Hispanic Scholar	March 2018
	Recognition of top national performers of Latin descent on the SAT	
	National Merit Scholar	March 2018
	Recognition of top national performers on the SAT	